CODING THEORY

ANDREW WOOD

Contents

1.	Introduction to Codes	1
2.	Vector Spaces over Finite Fields	3
3.	Linear Codes	7
4.	The Dual Code and Parity-Check Matrices	8
5.	Decoding	11
6.	Coding Bounds	13
7.	Perfect Codes	17
8.	Reed-Solomon Codes	18

1. Introduction to Codes

In this set of notes, we will describe error-correcting codes. The basic problem is to choose a $C \subseteq A^n$ such that we can detect and then correct errors without having too much redundancy.

Definition 1.1. A *q-ary alphabet* is a set of q distinct symbols. A *word* over an alphabet is a finite sequence of symbols from the alphabet. A *q-ary code* C is a set of words over a q-ary alphabet. The words in a code are called *codewords*. The *length* of a codeword is the number of symbols in it. If all codewords in C have a fixed length n, we say that C is a *(block) code of length* n.

Note. Set A is the *alphabet* of length q, ie, $|A| = q$. For $A^n = \{(a_1, \ldots, a_n) \mid a_i \in A\}$, take $C \subseteq A^n$ to be the *set of words* over the alphabet A. The *information rate* is

$$\frac{\log_q |C|}{n} = \frac{\log_{|A|} |C|}{n}.$$

For coding theory, we need to make some assumptions about the way errors occur. For a binary code, we assume that there is some probability p with $0 \leq p \leq 1$ so that a 0 gets changed to a 1 with probability p, and a 1 gets changed to a 0 with probability p. Clearly, the probability that a 0 or 1 remains unchanged is $1 - p$. Moreover, we assume that errors are *independent* in the statistical sense. Such a channel is known as the *binary symmetric channel*.

Definition 1.2. Consider a set of words of fixed length n over some alphabet. The *Hamming distance* between two words $\mathbf{u}$ and $\mathbf{v}$ is the number of places in which they differ. This distance is denoted by $d_h(\mathbf{u}, \mathbf{v})$ and given by the function $d_h : A^n \times A^n \to \{0, 1, \ldots, n\} \subseteq \mathbb{R}$.

Proposition 1.3. *Let C be a set of words of fixed length over some alphabet (that is, $C = A^n$). Then, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in C$, the Hamming distance satisfies the following properties:*

(1) $d_H(\mathbf{u}, \mathbf{v}) \geq 0$ *with equality if and only if* $\mathbf{u} = \mathbf{v}$*;*
(2) $d_H(\mathbf{u}, \mathbf{v}) = d_H(\mathbf{v}, \mathbf{u})$*; and*

(3) $d_H(\mathbf{u},\mathbf{v})+d_H(\mathbf{v},\mathbf{w}) \geq d_H(\mathbf{u},\mathbf{w})$. *This is known as the* Triangle Inequality.

Proof. By definition, $d_H(\mathbf{u},\mathbf{v}) \geq 0$. Furthermore,

$$d(\mathbf{u},\mathbf{v}) = 0 \iff |\{i \mid u_i \neq v_i\}| = 0 \iff \mathbf{u} = \mathbf{v}$$

thus proving (1).

We get that

$$|\{i \mid u_i \neq v_i\}| = |\{i \mid v_i \neq u_i\}| \iff d(\mathbf{u},\mathbf{v}) = d(\mathbf{v},\mathbf{u})$$

thus proving (2).

If $\mathbf{u} = \mathbf{v}$, we get equality and if $\mathbf{u} \neq \mathbf{v}$ we get inequality thus proving (3). □

Note. This shows that the Hamming distance is an example of a *metric*.

Definition 1.4. Let $\mathbf{u}$ be a word over the alphabet $\{0, 1, \ldots, q\}$. The *weight* of $w(\mathbf{u})$ is the number of non-zero components of $\mathbf{u}$.

Definition 1.5. The *distance of a code* C, denoted $d(C)$, is the minimum distance between any pair of distinct codewords. That is to say,

$$d(C) = \min_{\mathbf{u}\neq\mathbf{v}\in C} \{d(\mathbf{u},\mathbf{v})\}.$$

Note. Nearest Neighbour Decoding: Given $x \in A^n$ output $u \in C$ such that $d(u,x) = \min\{d(v,x) \mid v \in C\}$. We get

$$\mathbb{P}(\mathbf{y} \text{ received} \mid \mathbf{x} \text{ sent}) = p^d(1-p)^{n-d} = (1-p)^n\left(\frac{p}{1-p}\right)^d$$

where $d = d_H(\mathbf{y},\mathbf{x})$ and n is the code length. For $0 \leq p < \frac{1}{2}$, this is maximised precisely for the word(s) $\mathbf{x}$ for which $d = d_H(\mathbf{y},\mathbf{x})$ is minimum, that is, for the word(s) $\mathbf{x}$ nearest to $\mathbf{y}$ under the Hamming distance. That is, $\frac{p}{1-p} < 1$ if $p < \frac{1}{2}$.

Definition 1.6. Let $\mathbf{u} = \langle u_1, u_2, \ldots, u_n\rangle$ and $\mathbf{v} = \langle v_1, v_2, \ldots, v_n\rangle$ be two *binary* codewords. Then $\mathbf{u} \cap \mathbf{v} = \langle u_1v_1, u_2v_2, \ldots, u_nv_n\rangle$.

Proposition 1.7. *Let* $\mathbf{u}$ *and* $\mathbf{v}$ *be two* binary *codewords, then*

$$d_H(\mathbf{u},\mathbf{v}) = w(\mathbf{u}) + w(\mathbf{v}) - 2w(\mathbf{u}\cap\mathbf{v}).$$

Proof. Consider the pair u_i, v_i in $\mathbf{u},\mathbf{v}$. If $u_i \neq v_i$ we have only one case up to isomorphism. Without loss of generality, suppose $u_i = 0$ and $v_i = 1$. Then $w(\mathbf{u})$ for index i gives 0 and $w(\mathbf{v})$ at i gives 1; clearly $\mathbf{u}\cap\mathbf{v}$ yields 0 as well. Overall, we get 1 which is what we expect.

Now, suppose $u_i = v_i$. Then for RHS we get 0 for both cases which is what we expect and thus we are done. □

Theorem 1.8. *Let* $C \subseteq A^n$ *be a code.*

(1) *If* $d(C) \geq s+1$*, then* C *can detect up to* s *errors in any codeword. In other words,* C *is a* s*-error detecting code.*
(2) *If* $d(C) \geq 2t+1$*, then* C *can correct up to* t *errors in any codeword. In other words,* C *is a* t*-error correcting code.*

Proof. Suppose $d(C) \geq s+1$. Suppose a codeword x is transmitted and s or fewer errors are introduced. Then the received vector cannot be different and so the errors can be detected, thus proving (1).

Take $t = \left\lfloor \frac{d(C)-1}{2} \right\rfloor$. We claim that if $x \in A^n$ has at most t errors then we can recover $c \in C$ with $d(x,c) \leq t$ uniquely.

If c_1, c_2 both satisfy $d(x, c_1), d(x, c_2) \leq t$ then

$$\begin{aligned} & d(C) \leq d(c_1, c_2) \leq d(x, c_1) + d(x, c_2) \leq 2t \\ \implies \quad & d(C) \leq d(C) - 1 \end{aligned}$$

a contradiction. Thus, we recover a unique $c \in C$ which proves (2). □

Proposition 1.9. *Let C be a code with minimal distance $d(C) = 2m$. If we receive a message with at most m errors we can tell how many errors it has.*

Proof. Suppose we receive a message $\mathbf{u}$ with at most m errors. We consider two cases, one where $d(\mathbf{u}, c) = m$ and when $d(\mathbf{u}, c) < m$. Then we claim we can recover $c \in C$ uniquely if $d(\mathbf{u}, c) < m$.

Suppose that $d(\mathbf{u}, c) < m$. If both $c_1 \neq c_2 \in C$ satisfy $d(\mathbf{u}, c_1), d(\mathbf{u}, c_2) < m$, then by the triangle inequality we get that $d(C) = 2m \leq d(c_1, c_2) \leq d(\mathbf{u}, c_1) + d(\mathbf{u}, c_2) < 2m$ and so we get that $d(C) < d(C)$ (a contradiction). So c must be unique, proving the claim.

Whether or not uniqueness holds for $d(\mathbf{u}, c) = m$ is irrelevant, for we can tell precisely how many errors $\mathbf{u}$ contains by computing $d(\mathbf{u}, c)$ for every $c \in C$ and taking the minimum. This is because if the minimum is less than m, we know it to be unique (by the claim) and so the number of errors of $\mathbf{u}$. Otherwise, we know the number of errors of $\mathbf{u}$ to be m. In either case, we can tell how many errors it has. □

2. Vector Spaces over Finite Fields

Definition 2.1. A *field* $\mathbb{F}$ is a set of elements together with two binary operations $+$ and $\cdot$ satisfying the following properties:

(1) $\mathbb{F}$ is *closed* under $+$ and $\cdot$, that is, for all $a, b \in \mathbb{F}$,
$$a + b \in \mathbb{F} \text{ and } a \cdot b \in \mathbb{F}.$$
(2) The *commutative* laws hold for $+$ and $\cdot$, that is, for all $a, b \in \mathbb{F}$,
$$a + b = b + a \text{ and } a \cdot b = b \cdot a.$$
(3) The *associative* laws hold for $+$ and $\cdot$, that is, for all $a, b, c \in \mathbb{F}$,
$$(a + b) + c = a + (b + c) \text{ and } a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$
(4) The *distributive* law holds. In other words, for all $a, b, c \in \mathbb{F}$,
$$a \cdot (b + c) = a \cdot b + a \cdot c.$$
(5) There is an element $0 \in \mathbb{F}$ such that, for all $a \in \mathbb{F}$, $a + 0 = a$. The element 0 is called the *additive identity* of $\mathbb{F}$.
(6) There is an element $1 \in \mathbb{F}$ such that, for all $a \in \mathbb{F}$, $a \cdot 1 = a$. The element 1 is called the *multiplicative element* of $\mathbb{F}$.
(7) For each $a \in \mathbb{F}$, there is an element $-a \in \mathbb{F}$ such that $a + (-a) = 0$. The element $-a$ is called the *additive inverse* of a.
(8) For each $a \neq 0 \in \mathbb{F}$, there is an element $a^{-1} \in \mathbb{F}$ such that $a \cdot a^{-1} = 1$. The element a^{-1} is called the *multiplicative inverse* of a.

Definition 2.2. Let m be a fixed positive integer. Integers a and b are *congruent* (*modulo* m), denoted $a \equiv b \mod m$, if $a - b$ is divisible by m; that is, if $a = km + b$ for some integer k.

Note. Every integer, when divided by m has a *principal* remainder in $\mathbb{Z}_m$. We get that $\mathbb{Z}_m$ is a *ring* and satisfies the first 7 axioms of a field for all m. It satisfies the last axiom if and only if m is prime, in which case $\mathrm{GF}(m) = \mathbb{Z}_m$.

Proposition 2.3. *If $a \equiv a' \mod m$ and $b \equiv b' \mod m$, then*

(1) $a + b \equiv a' + b' \mod m$, *and*
(2) $ab \equiv a'b' \mod m$.

Proof. It follows that $m \mid (a - a')$ and $m \mid (b - b')$ which certainly implies that $m \mid (a + b - a' - b') \iff a + b \equiv a' + b' \mod m$ thus proving (1).

Furthermore,

$$m \mid (a - a')(b - b') \iff m \mid (ab + a'b' - ab' - a'b) \iff m \mid (ab - a'b' + a'(b' - b) + b'(a' - a))$$

and so it follows that $m \mid (ab - a'b') \iff ab \equiv a'b' \mod m$ thus proving (2). □

Theorem 2.4. *$\mathbb{Z}_m$ is a field if and only if m is a prime number.*

Proof. We know that $\mathbb{Z}_m$ satisfies the first 7 field axioms for all m and so we need only prove that $\mathbb{Z}_m$ satisfies (8) if and only if m is prime.

($\Longrightarrow$) Firstly, suppose that $\mathbb{Z}_m$ is a field. Then for each $a \neq 0 \in \mathbb{Z}_m$, there is an element $b \in \mathbb{Z}_m$ such that $ab \equiv 1 \mod m$. That is to say, $m \mid (ab - 1)$ which means that there exists $k \in \mathbb{Z}$ such that $mk = ab - 1 \iff ab - mk = 1 \iff \gcd(a, m) = 1$. Since a is arbitrary, it follows that m is prime.

($\Longleftarrow$) Now, suppose m is prime. Then for all $a \in \mathbb{Z}_m : a \neq 0$, $\gcd(a, m) = 1$ which means that there exists $b \in \mathbb{Z}_m$ and $k \in \mathbb{Z}$ such that $ab + km = 1$. This implies that $ab \equiv 1 \mod m$, which proves the theorem. □

Theorem 2.5. *Let m be a positive integer. Then there exists a field with m elements if and only if m is a prime power (that is, $m = p^n$ for some prime p). Moreover, if m is a prime power, then this field is unique up to isomorphism.*

Proof. □

Note. For a prime power m, the field with m elements is denoted by $\mathrm{GF}(m)$, where GF stands for *Galois field*. If m is prime, then $\mathrm{GF}(m) = \mathbb{Z}_m$. If m is not prime, then $\mathrm{GF}(m)$ and $\mathbb{Z}_m$ are *very different structures*.

Definition 2.6. Let $\mathbb{F}$ be a field and V be a set. Elements of $\mathbb{F}$ are called *scalars* and elements of V are called *vectors*. Assume we have a binary operation $+$ on V, and an operation called *scalar operation* that associates with every $a \in \mathbb{F}$ and $\mathbf{v} \in V$ an element $a\mathbf{v}$. Then V is a *vector space* over $\mathbb{F}$ if, for all $a, b \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in V$, the following properties are satisfied:

(1) $+$ is closed, associative, commutative, has an additive identity (denoted $\mathbf{0}$), and each $\mathbf{u} \in V$ has an additive inverse (denoted $-\mathbf{u}$);
(2) $a\mathbf{u} \in V$;
(3) $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$;
(4) $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$;
(5) $(ab)\mathbf{u} = a(b\mathbf{u})$; and
(6) $1\mathbf{u} = \mathbf{u}$, where 1 is the multiplicative identity of $\mathbb{F}$.

Proposition 2.7. *Let $\mathbb{F}$ be a field, and let V be the set of all ordered n-tuples of elements of $\mathbb{F}$ with addition defined by*

$$(u_1, \ldots, u_n) + (v_1, \ldots, v_n) = (u_1 + v_1, \ldots, u_n + v_n)$$

and scalar multiplication defined by

$$a(u_1, \ldots, u_n) = (au_1, \ldots, au_n).$$

Then V is a vector space over $\mathbb{F}$.

Proof. Since $\mathbb{F}$ is a field, there exists an additive identity 0 such that

$$\mathbf{0} = (0, \ldots, 0)$$

is the additive identity for V. $+$ being closed, associative and commutative also follows from $\mathbb{F}$ being a field. Furthermore, we get the additive inverse of each and so

$$(u_1, \ldots, u_n) + (-u_1, \ldots, -u_n) = \mathbf{0}$$

satisfies (1).

(2) is satisfied by scalar multiplication.
(3) is satisfied by scalar multiplication over addition.
(4) is satisfied by addition and scalar multiplication.
(5) is satisfied by scalar multiplication with associativity.
(6) is satisfied by scalar multiplication using multiplicative identity of $\mathbb{F}$. □

Note. The vector space of n-tuples over the finite field $\mathrm{GF}(q)$ is denoted by $V(n, q)$, which has q^n many vectors.

Definition 2.8. Let V be a vector space over the field $\mathbb{F}$ and let $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ be a subset of vectors of V. A *linear combination* of X is a vector of the form

$$a_1\mathbf{x}_1 + \ldots + a_n\mathbf{x}_n,$$

where $a_1, \ldots, a_n \in \mathbb{F}$.

The *span* of X, denoted $\langle X \rangle$ or $\mathrm{Span}(X)$, is the set of all linear combinations of vectors in X.

A *subspace* of V is a subset of V that is a vector space in its own right.

Theorem 2.9. *A non-empty subset C of a vector space V over a field $\mathbb{F}$ is a subspace of V if and only if C is closed under addition and scalar multiplication, that is,*

(1) *if $\mathbf{x}, \mathbf{y} \in C$, then $\mathbf{x} + \mathbf{y} \in C$, and*
(2) *if $a \in \mathbb{F}$ and $\mathbf{x} \in C$, then $a\mathbf{x} \in C$.*

Proof. ($\Longrightarrow$) Suppose that C is a subspace of V and that $\mathbf{x}, \mathbf{y} \in C$. Then $\mathbf{x} + \mathbf{y} = (x_1 + y_1, \ldots, x_n + y_n)$. Since $+$ is closed, we get that $x_1 + y_1, \ldots, x_n + y_n \in \mathbb{F}$ which implies that $\mathbf{x} + \mathbf{y} \in C$ thereby proving (1). To prove (2), we simply take the second axiom of a vector space.

($\Longleftarrow$) Suppose that C is closed under addition and scalar multiplication. It can be checked that C upholds the 6 vector space axioms. □

Proposition 2.10. *Let V be a vector space and let X be a subset of V. Then $\langle X \rangle$ is a subspace of V.*

Proof. Since $\langle X \rangle$ is every possible linear combination of vectors in X, we get that it must be closed under addition and scalar multiplication. Thus, $\langle X \rangle$ must be a subspace of V. □

Definition 2.11. A subset $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ of a vector space is *linearly independent* if the only solution to

$$a_1\mathbf{x}_1 + \ldots + a_n\mathbf{x}_n = \mathbf{0}$$

is $a_1 = a_2 = \ldots = a_n = 0$. Otherwise, X is *linearly dependent.*

Proposition 2.12. *A subset X of a vector space is linearly independent if and only if no vector in X can be written as a linear combination of the others.*

Proof. A subset X of a vector space is linearly independent if and only if the only solution to

$$a_1\mathbf{x}_1 + \ldots + a_n\mathbf{x}_n = \mathbf{0}$$

is $a_1 = a_2 = \ldots = a_n = 0$ which happens if and only if

$$a_1\mathbf{x}_1 + \ldots + a_n\mathbf{x}_n = \mathbf{x}_i$$

has no solution since $a_i = -1$ and yet we know $a_i = 0$ for arbitrary i. □

Definition 2.13. A *basis* for a vector space V is a linearly independent subset of V that spans V. In other words, X is a basis for V if X is linearly independent and $\langle X \rangle = V$.

Note. V is *finite dimensional* if it has a basis with a finite number of elements. The number of vectors in a basis of a finite dimensional vector space V is called the *dimension* of V.

The *standard* basis of $V(n, q)$ is the set

$$\{(1, 0, \ldots, 0), (0, 1, \ldots, 0), \ldots, (0, 0, \ldots, 1)\}$$

of dimension n.

Proposition 2.14. *The standard basis of $V(n, q)$ is indeed a basis.*

Proof. It is easily checkable that for any non-zero entry of a vector we can multiply said entry by the corresponding identity vector. □

Proposition 2.15. *If V is a finite dimensional vector space, then every basis of V has the same cardinality. In other words, if X and Y are bases of V, then X and Y both contain the same number of vectors.*

Proof. Since $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ and $Y = \{\mathbf{y}_1, \ldots, \mathbf{y}_m\}$ are bases,

$$\mathbf{y}_i = a_1\mathbf{x}_1 + \ldots + a_n\mathbf{x}_n$$

and

$$\mathbf{x}_j = b_1\mathbf{y}_1 + \ldots + b_m\mathbf{y}_m$$

for all i, j and $\langle X \rangle = \langle Y \rangle = V$. Without loss of generality, and for sake of contradiction, suppose that $n < m$. It follows that there exists some vector in Y that is not in X (yet of course in the span of X), which we denote $\mathbf{y}_i$. We then express all other $\mathbf{x}_j \in X$ as linear combinations of vectors in Y giving us the linearly independent set $\overline{X}$ that is also a basis. We then define the linearly dependent set $Z = \overline{X} \cup \{\mathbf{y}_i\}$ whereby $\mathbf{y}_i \notin \overline{X}$. Since Z is linearly dependent, we can find a $\mathbf{y}_j$ with not all non-zero constants such that we have a linear combination of other vectors in Y to get $\mathbf{y}_j$, a contradiction. □

Proposition 2.16. *Let V be a k-dimensional vector space and let X be a subset of V. Then X is a basis of V if and only if X is linearly independent and X has k vectors.*

Proof. X is a basis of V if and only if X is linearly independent and spans V. Furthermore, the standard basis is k-dimensional and thus any basis must have k vectors.

Now, suppose $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ has k vectors that are linearly independent. That is,

$$a_1\mathbf{x}_1 + \ldots + a_k\mathbf{x}_k = \mathbf{0}$$

has only the trivial solution. Then each assignment of $a_1, \ldots, a_k$ gives a unique vector in V. For any vector in V, we must choose k many constants from the field. This implies that the linear combinations of X form a bijection from the span of X to V. Thus, $\mathrm{Span}(X) = V$. □

Note. This proof relies upon the fact that we are dealing with finite fields.

3. Linear Codes

What makes a good code? When encoding messages to codewords, you add extra symbols. The number of extra symbols is called the *redundancy* of the code. You can make a code detect and correct a lot of errors by adding lots of redundancy, but then it's going to take a lot of time to transmit a reasonable amount of information. The basic problem in coding theory is to find codes that detect and correct as many errors as possible with a minimum of redundancy. Recall that $V(n,q) = \mathrm{GF}(q)^n$.

Definition 3.1. If q is a prime power, then the set of all words of length n from the q-ary alphabet $\{0, 1, 2, \ldots, q-1\}$ can be regarded as the vector space $V(n,q)$. Such a code is a *linear code* over $\mathrm{GF}(q)$ if the set of all codewords is a *subspace* of $V(n,q)$.

Note. $\mathrm{GF}(q)$ is a field with q elements. When q is a prime power, $\mathrm{GF}(q)$ is the field with q elements up to isomorphism. If q is prime, then $\mathrm{GF}(q) = \mathbb{Z}_q$.

Theorem 3.2. *If C is a linear code, then $d(C)$ is equal to the smallest weight of all non-zero codewords of C.*

Proof.

$$\begin{aligned} & d(\mathbf{u},\mathbf{v}) = |\{i \mid u_i \neq v_i\}| \text{ and } d(\mathbf{u}-\mathbf{v},\mathbf{0}) = |\{i \mid u_i - v_i \neq 0\}| \\ \implies \quad & d(\mathbf{u},\mathbf{v}) = d(\mathbf{u}-\mathbf{v},\mathbf{0}) \quad \text{due to linearity} \\ \implies \quad & d(C) = \min_{\mathbf{u}\neq\mathbf{v}\in C}\{d(\mathbf{u},\mathbf{v})\} = \min_{\mathbf{w}\in C:\mathbf{w}\neq\mathbf{0}}\{d(\mathbf{w},\mathbf{0})\} = w(C) \end{aligned}$$

□

We have three parameters of a linear code: As well as having *length* n (number of symbols in a codeword) and *distance* d, linear codes have a *dimension* k. A linear code with these three parameters is called an $[n,k,d]-$(*linear*) *code*. Do note that $\frac{k}{n}$ is the *information rate*.

Definition 3.3. A *generator matrix* for a linear code C is a matrix whose rows form a basis of C. Thus a generator matrix G for C defines C; in particular, $\mathbf{v} \in C$ precisely if $\mathbf{v}$ is a linear combination of the rows of G.

Note. A generator matrix G of a code C is in *standard coding form* if it is of the form $[I \mid A]$ where I is the identity matrix. If a generator matrix of C is in standard coding form it encodes messages by adding extra symbols. If C has dimension k, then the first k columns of G in standard coding form is I_k. Multiplying a message of length k by G means that the first k symbols of the resulting codeword is identical to the message. Do not that not all linear codes have a generator matrix in standard coding form.

Proposition 3.4. *Let B be a basis of a subspace W of $V(n,q)$. Then each vector in W can be uniquely written as a linear combination of the vectors in B. Consequently, a subspace of dimension k has q^k vectors.*

Proof. We get $B = \{b_1, \ldots, b_k\}$ such that $|B| = k$ and $W = \{\lambda_1 b_1 + \ldots \lambda_k b_k \mid \lambda_1, \ldots, \lambda_k \in \mathrm{GF}(q)\}$. We get a one-to-one correspondence between $V(k,q)$ and the vectors in the subspace W. Since $|V(k,q)| = q^k$, so must W. □

Corollary 3.5. *A generator matrix of a linear code encodes distinct messages to distinct codewords.*

Proof. G is a generator matrix if the rows form a basis. If $aG = bG$, then $(a-b)G = \mathbf{0}$. For G with rows $\mathbf{w}_1, \ldots, \mathbf{w}_k$ and $a = (a_1, \ldots, a_k)$ and $b = (b_1, \ldots, b_k)$ we get $(a_1 - b_1)\mathbf{w}_1 + \ldots + (a_k - b_k)\mathbf{w}_k = \mathbf{0}$. Since the rows are linearly independent and

non-zero, this occurs precisely when $a_i - b_i = 0$ for all i, which implies $a_i = b_i$ and by extension $a = b$ thus proving the result. □

Proposition 3.6. *A non-empty subset W of $V(n, 2)$ is a subspace if and only if, for all $\mathbf{u}, \mathbf{v} \in W$, we have $\mathbf{u} + \mathbf{v} \in W$.*

Proof. A non-empty subset W of $V(n, 2)$ is a subspace if and only if, W is closed under scalar multiplication and closed under vector addition. Since scalar multiplication of vectors in W either give $\mathbf{0}$ or itself, and since $\mathbf{u} + \mathbf{u} = \mathbf{0}$, we get get both properties are satisfied. Hence, we need only have closure under addition which proves the result. □

Some advantages of linear codes:

- Easy to encode/decode with a generator matrix;
- can compute with distance by computing the minimum weight;
- and there are decoding algorithms that use linearity.

Theorem 3.7. *Let C be a q-ary linear code and $C_\lambda^{[i]} = \{c \in C \mid c_i = \lambda\}$ for position i and $\lambda \in GF(q)$. If $C_\lambda^{[i]}$ is non-empty, then $\phi : C_0^{[i]} \to C_\lambda^{[i]}$ is a bijection.*

Proof. Suppose that $C_\lambda^{[i]} \neq \varnothing$. Then there exists $c_1, c_2 \in C_\lambda^{[i]}$ such that $c_1 - c_2 = x \iff c_1 = x + c_2$ for some $x \in C_0^{[i]}$. The map $\phi : C_0^{[i]} \to C_\lambda^{[i]}$ defined $\phi : x \mapsto x + y$ for some $y \in C_\lambda^{[i]}$ is then one-to-one and onto (since it is invertible). Thus, ϕ is a bijection. □

Note. It therefore follows that if every symbol in a q-ary alphabet is at position i for some codeword then $\frac{1}{q}$ is the probability it is some arbitrary symbol λ. This in fact gives us the lower bound as $\frac{1}{q}$.

4. The Dual Code and Parity-Check Matrices

Definition 4.1. Recall that, for two vectors $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$, the *dot product* is defined by

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \ldots + u_n v_n.$$

Furthermore, we say that $\mathbf{u}$ and $\mathbf{v}$ are *orthogonal* if $\mathbf{u} \cdot \mathbf{v} = 0$.

Note. Over $\mathbb{R}$, recall that we defined $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$. For some fields, such as the reals, the dot product is an example of an *inner product*. However, for other fields, such as GF(2), it is not an inner product. For it to be an inner product, $\mathbf{v} \cdot \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$. However, for GF(2), we have $(1, 1) \cdot (1, 1) = 0$ and yet $(1, 1) \neq (0, 0)$.

Proposition 4.2. *If W is a subspace of a vector space V, then the set*

$$W^\perp = \{\mathbf{v} \in V \mid \mathbf{v} \cdot \mathbf{w} = 0 \textit{ for all } \mathbf{w} \in W\}$$

is a subspace of V. This subspace is called the orthogonal subspace *of W.*

Proof. Let $\mathbf{u}, \mathbf{v} \in W^\perp$ and $\lambda_1, \lambda_2 \in \mathbb{F}$ be arbitrary. Suppose $\mathbf{w} \in W$. Then $\mathbf{u} \cdot \mathbf{w} = 0$ and $\mathbf{v} \cdot \mathbf{w} = 0$ which certainly implies $(\lambda_1 \mathbf{u}) \cdot \mathbf{w} + (\lambda_2 \mathbf{v}) \cdot \mathbf{w} = 0$. It follows that $(\lambda_1 \mathbf{u} + \lambda_2 \mathbf{v}) \cdot \mathbf{w} = 0$ which proves the result. □

Note. If C is a subspace of $V(n, q)$, then C is a linear code. The orthogonal subspace $C^\perp$ of C is called the *dual code* of C.

Proposition 4.3. *Let W be a subspace of the vector space V. Then*

(1) *the dimension of V is the sum of the dimensions of W and $W^\perp$, and*
(2) $\left(W^\perp\right)^\perp = W$.

Proof. (1) is a direct consequence of the Rank-Nullity Theorem.

Recall that

$$W^{\perp} = \{\mathbf{v} \in V \mid \mathbf{v} \cdot \mathbf{w} = 0 \text{ for all } \mathbf{w} \in W\}.$$

It therefore follows that

$$\left(W^{\perp}\right)^{\perp} = \left\{\mathbf{v} \in V \;\middle|\; \mathbf{v} \cdot \mathbf{w} = 0 \text{ for all } \mathbf{w} \in W^{\perp}\right\}.$$

Since, for $\mathbf{v} \in W^{\perp}$, we get that $\mathbf{v} \cdot \mathbf{w} = 0$ for all $\mathbf{w} \in W$, this is equivalent to saying, for $\mathbf{v} \in W$, we get that $\mathbf{v} \cdot \mathbf{w} = 0$ for all $\mathbf{w} \in W^{\perp}$. Yet, the latter is also the condition for $\mathbf{v} \in \left(W^{\perp}\right)^{\perp}$, so $W \subseteq \left(W^{\perp}\right)^{\perp}$. By (1), we it follows that the dimensions of W and $\left(W^{\perp}\right)^{\perp}$ are equal which proves (2). □

Definition 4.4. The *parity-check* matrix H for a linear code C is a generator matrix for $C^{\perp}$.

Note. Parity-check matrices are essential for *decoding*.

Proposition 4.5. *Let C be a linear code of length n and dimension k, and suppose that H is a parity-check for C. Let C be a subspace of $V(n,q)$ and let $\mathbf{v} \in V(n,q)$. Then $\mathbf{v} \in C$ if and only if $\mathbf{v}H^T = \mathbf{0}$.*

Proof. $H^T = h_1^T \dots h_{n-k}^T$, where h_i forms a basis for $C^{\perp}$. Therefore:

$$\begin{aligned} \mathbf{v}H^T = \mathbf{0} &\iff \mathbf{v}h_i = 0 \\ &\iff \mathbf{v} \cdot h = 0 \quad \forall h \in C^{\perp} \\ &\iff \mathbf{v} \in \left(C^{\perp}\right)^{\perp} = C. \end{aligned}$$

□

Note. Codes used in practice have a large number of codewords. Keeping a list of all codewords is impractical (even on a computer). Parity-check matrices provide a quick way of checking that a received word is indeed a codeword.

Giving a code as the solution set of a system of linear equations is the same as giving the parity check matrix of the code.

Definition 4.6. Let H be a parity check for the linear binary code C. We say that H has *low density*, if the entries of H are 'mostly' zeroes.

Theorem 4.7. *Let C be a linear code of length n and dimension k. If $G = [I_k \mid A_{k\times(n-k)}]$ is a generator matrix for C in standard coding form, then*

$$H = [-A^T_{(n-k)\times k} \mid I_{n-k}]$$

is a parity-check matrix for C.

Proof. Suppose that G is a generator matrix for C. Note that G is a $k \times n$ matrix, and that H is an $(n-k)\times n$ matrix. So, $G \cdot H^T$ is valid, which results in a $k \times (n-k)$ matrix.

Now, we claim that $G \cdot H^T = \mathbf{0}_{k\times(n-k)}$ if and only if H is a parity-check matrix for C. For our purposes, however, we need only prove the forward direction (see Tutorial 1 Question 5 for full proof of this fact). Clearly, it is obvious to notice that H has rank $n-k$ which gives us H has linearly independent rows. Furthermore, if $\mathbf{u}$ is a row of G and $\mathbf{v}$ is a row of H, then their dot product must be zero (since $G \cdot H^T$ is the zero matrix). Thus, the row space of G and row space of H must be orthogonal which implies H is a parity-check matrix for C, which proves our claim (in the forward direction).

Now, we need only show that $G \cdot H^T$ gives the zero matrix. We get that

$$H^T = \begin{bmatrix} -A_{k\times(n-k)} \\ I_{n-k} \end{bmatrix}$$

which then implies

$$G \cdot H^T = [I_k \mid A_{k\times(n-k)}] \cdot \begin{bmatrix} -A_{k\times(n-k)} \\ I_{n-k} \end{bmatrix} = -A_{k\times(n-k)} + A_{k\times(n-k)} = \mathbf{0}_{k\times(n-k)}$$

thus proving H is a parity-check matrix for C. □

Note. If C is binary, then $-A^T_{(n-k)\times k} = A^T_{(n-k)\times k}$.

Proposition 4.8. *Let H be a parity-check matrix for a binary linear code C. If no column of H consists entirely of zeroes and no two columns of H are the same, then $d(C) \geq 3$, and so C is a single-error correcting code.*

Proof. We will provide a proof by contrapositive. That is, supposing that $d(C) \in \{1, 2\}$ and concluding that either H has a column of zeroes or two columns of H are the same.

Let $h_1, \ldots, h_{n-k}$ be the rows of H, and let $c \in C$. Suppose that the hamming weight of c is 1. Then $d(C) = 1$, and c is a vector with its ith coordinate as 1. We know that $ch_m = 0$ for all m, yet this implies that a zero column exists. Now, suppose instead that the hamming weight of c is 2. Then c has a 1 at the ith and jth coordinate. We then know that $ch_m = 0$ which implies that the ith and jth position in the rows of H must be equal. That is, the ith and jth columns of H are equal. □

Proposition 4.9. *Let H be a parity-check matrix for a linear code C over $GF(q)$. Then*

(1) *$d(C) = 1$ if and only if there is a zero column in H; and*
(2) *$c \in C$ has weight 2 if and only if there are 2 linearly dependent columns of H.*

Consequently, if H has no zero column and no two columns of H are linearly dependent, then $d(C) \geq 3$.

Proof. Let $h_1, \ldots, h_{n-k}$ be the rows of H, and let $c \in C$ such that $w(c) = 1$. Then c is a vector with its ith coordinate as $\lambda \in \mathrm{GF}(q)$. We know that $ch_m = 0$ for all m, which happens if and only if a zero column exists which proves (1).

Now, suppose $c \in C$ has weight 2. This happens if and only if c has λ_i, λ_j at the ith and jth positions respectively. Then $\lambda_i \mathbf{x}_i + \lambda_j \mathbf{x}_j = 0$ where $\mathbf{x}_i, \mathbf{x}_j$ are the columns at the ith and jth positions. So, they are linearly dependent which proves (2). □

Theorem 4.10. *A linear code C with parity-check matrix H has minimum distance greater than or equal to d if no set of $d-1$ columns of H are linearly dependent. $d(C) = d$ if, in addition, there are d linearly dependent columns.*

Proof. Suppose no set of $d-1$ columns of H are linearly dependent. Suppose $c \in C$ has $w(c) \leq d-1$. Then for $w(c)$ many columns at the non-zero positions of c must have their sum be 0. Yet, this implies a linear dependent set of $d-1$ columns of H. Thus, $d(C) \geq d$. Consequently, $d(C) = d$ if there are d many linearly dependent columns. □

Note. Furthermore, $d+k \leq n+1$. We know the rank of H to be $n-k$. Since no set of $d-1$ columns of H are linearly independent, it follows that $d-1 \leq \mathrm{rank}(H) = n-k$ which rearranging yields $d + k \leq n + 1$. This is called a singleton bound.

Definition 4.11. A *permutation* of a set S is a one-to-one mapping f from S into itself.

Note. A map is a *bijection* if it is both one-to-one and onto.

Definition 4.12. Two q-ary codes C and C', not necessarily linear, are *equivalent* if one can be obtained from the other by any sequence of operations of the following types:

(1) permuting the positions of the code; and
(2) permuting the symbols in some fixed position.

Proposition 4.13. *If C and C' are equivalent codes, then there is a bijection $f : C \to C'$ such that, for all $\mathbf{u}, \mathbf{v} \in C$,*

$$d(\mathbf{u}, \mathbf{v}) = d(f(\mathbf{u}), f(\mathbf{v})).$$

That is, the map preserves distance which is called isometry.

Proof. Let $\mathbf{u}, \mathbf{v} \in C$ such that $\mathbf{u} = (u_1, \ldots, u_n)$ and $\mathbf{v} = (v_1, \ldots, v_n)$. We get that

$$d(\mathbf{u}, \mathbf{v}) = |\{i \mid u_i \neq v_i\}|$$
$$d(\mathbf{u}, \mathbf{v}) = n - |\{i \mid u_i = v_i\}|.$$

Now, suppose we permute the symbols in some fixed positions. We get that if $u_i = v_i$, then u_i and v_i will be mapped to the same element (since they are the same elements and same positions). Therefore,

$$d(f(\mathbf{u}), f(\mathbf{v})) = n - |\{i \mid f(u_i) = f(v_i)\}|.$$

Now, suppose that we permute the positions by swapping elements. Clearly, if $u_i = v_i$ at position i, and position i moves to position j then $f(u_i) = f(v_i)$. Thus, f preserves distance. □

Theorem 4.14. *Let C be a linear code and let G be a generator matrix for C. Let G' be a matrix that is obtained from G using matrix column and row operations. Then G' is a generator matrix for a linear code C' that is equivalent to C.*

Proof. □

Theorem 4.15. *By using matrix column and row operations, a generator matrix of a linear code C can be transformed into a generator matrix in standard coding form for an equivalent code C'.*

Proof. □

5. Decoding

Definition 5.1. Let W be a subspace of a vector space V and let $\mathbf{a} \in V$. The set

$$\{\mathbf{a} + \mathbf{w} \mid \mathbf{w} \in W\}$$

denoted by $\mathbf{a} + W$, is called a *coset* of W.

Lemma 5.2. *Let W be a subspace of a vector space V and let $\mathbf{a} \in V$. If $\mathbf{a} + W$ is a coset of W and $\mathbf{b} \in \mathbf{a} + W$, then $\mathbf{a} + W = \mathbf{b} + W$.*

Proof. We claim that $\mathbf{a} \in W$ if and only if $\mathbf{a} + W = W$. We know $\mathbf{a} \in W$ if and only if $-\mathbf{a} + \mathbf{w} \in W$ for all $\mathbf{w} \in W$. Then $\mathbf{w} \in \mathbf{a} + W$ for all $\mathbf{w} \in W$, which happens if and only if $\mathbf{a} + W = W$.

Now, let $\mathbf{a} \in V$ be arbitrary. Then if $\mathbf{b} \in \mathbf{a} + W$ we get $-\mathbf{a} + \mathbf{b} \in W$. Thus

$$\mathbf{a} + W = \mathbf{a} + -\mathbf{a} + \mathbf{b} + W = \mathbf{b} + W.$$

□

Proposition 5.3. *Let W be a subspace of a vector space V. Then the cosets of W partition V.*

Proof. Since the distinct cosets of W are disjoint, we get that they form a partition of V. □

Proposition 5.4. *Let W be a subspace of a vector space V, and let $\mathbf{a}, \mathbf{b} \in V$. Then $\mathbf{a}$ and $\mathbf{b}$ are in the same coset of W if and only if $\mathbf{a} - \mathbf{b} \in W$.*

Proof. $\mathbf{a}$ and $\mathbf{b}$ are in the same coset of W if and only if $\mathbf{b} \in \mathbf{a} + W$ which happens if and only if $\mathbf{b} - \mathbf{a} \in W$ which happens if and only if $\mathbf{a} - \mathbf{b} \in W$. □

Proposition 5.5. *If C is a k-dimensional subspace of $V(n, q)$, then every coset of C has exactly q^k vectors. Moreover, there are q^{n-k} cosets altogether.*

Proof. The order of $V(n, q)$ is q^n. Since C has q^k vectors, and cosets form a bijection, we get that each coset of C has q^k vectors. Since they must form a partition of the q^n vectors, we get that there are q^{n-k} vectors in total. □

Definition 5.6. For a linear code C with $p + 1$ codewords, a *(Slepian) standard array* is a way of displaying the cosets of C. It is constructed as follows. For the first row, list all codewords of C starting with $\mathbf{0}$:

$$\mathbf{0}, \mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p.$$

This is the coset $\mathbf{0} + C$. Now choose a vector $\mathbf{v}_1 \in V(n, q) \setminus C$ of minimum weight. There will typically be more than one choice of $\mathbf{v}_1$; no problem, it doesn't matter which one we choose. For the second row, list the coset $\mathbf{v}_1 + C$ as follows:

$$\mathbf{v}_1 + \mathbf{0}, \mathbf{v}_1 + \mathbf{u}_1, \mathbf{v}_1 + \mathbf{u}_2, \ldots, \mathbf{v}_1 + \mathbf{u}_p.$$

In the same way as before, choose a vector $\mathbf{v}_2$ of minimum weight in $V(n, q) \setminus (C \cup (\mathbf{v}_1 + C))$ and list $\mathbf{v}_2$. Repeat this process until all vectors in V are listed.

The elements $\mathbf{0}, \mathbf{v}_1, \mathbf{v}_2 \ldots$ are called the *coset leaders*.

We therefore have an algorithm. If we receive a message $\mathbf{u} = \mathbf{v} + c$ where $c \in C$ and $\mathbf{v}$ is a coset leader, locate $\mathbf{u}$ in row of $\mathbf{v}$. Then take c to be the row of $\mathbf{0}$ and column of $\mathbf{v}$.

Theorem 5.7. *Standard array decoding is nearest neighbour decoding.*

Proof. If we receive a vector $\mathbf{v}$ which is in the row $\mathbf{a} + C$, where $\mathbf{a}$ is the coset leader (and so has minimal weight in its coset), then $\mathbf{v} \in \mathbf{a} + C$. Then there exists some codeword $c \in C$, which we decode to using Standard array decoding, such that $\mathbf{v} = \mathbf{a} + c$. Therefore, $d(c, \mathbf{v}) = d(c - \mathbf{v}, \mathbf{0}) = w(c - \mathbf{v})$ and since $\mathbf{a} = c - \mathbf{v}$, it follows that $d(c, \mathbf{v}) = w(\mathbf{a})$ which is the minimal weight.

Suppose now that there exists some other codeword $c' \in C$ that we would decode to using nearest neighbour decoding. That is, we have $d(\mathbf{v}, c') < d(\mathbf{v}, c)$. Then $\mathbf{v} - c' = \mathbf{a}'$ and so we get $w(\mathbf{a}') < w(\mathbf{a})$. Hence, when setting up the Standard array, we would choose $\mathbf{a}'$ to be a coset leader before $\mathbf{a}$. So certainly $\mathbf{a}'$ is a coset leader. This implies that $\mathbf{v} \in \mathbf{a}' + C$, as well as $\mathbf{v} \in \mathbf{a} + C$. We get a contradiction by overlapping cosets, or breaking the assumption that $\mathbf{a}$ was of minimal weight in its coset. In either case, we have that nearest neighbour decoding decodes to c as well. □

Note that standard array decoding in practice is computationally infeasible, since we need q^n vectors in the standard array and search every time adding new coset.

Definition 5.8. Let C be a linear code with parity-check matrix H and let $\mathbf{u}$ be a vector in $V(n, q)$. The *syndrome* of $\mathbf{u}$ is the vector $\mathbf{u}H^T \in V(n - k, q)$.

Proposition 5.9. *Two vectors $\mathbf{u}$ and $\mathbf{v}$ are in the same coset of a linear code C if and only if $\mathbf{u}H^T = \mathbf{v}H^T$, where H is a parity-check matrix for C. In other words, $\mathbf{u}$ and $\mathbf{v}$ are in the same coset if and only if they have the same syndrome.*

Proof.

$$\begin{aligned} & \mathbf{u}H^T = \mathbf{v}H^T \\ \iff \quad & \mathbf{u}H^T - \mathbf{v}H^T = \mathbf{0} \\ \iff \quad & (\mathbf{u}-\mathbf{v})H^T = \mathbf{0} \\ \iff \quad & \mathbf{u}-\mathbf{v} \in C \\ \iff \quad & \mathbf{u}, \mathbf{v} \text{ in same coset.} \end{aligned}$$

□

Proposition 5.10. *Let C be a linear code with parity-check matrix H. The map $\Phi : V(n,q) \to V(n-k,q)$ defined $\Phi : \mathbf{u} \mapsto \mathbf{u}H^T$ is onto with $\ker \Phi = C$. Hence Φ^{-1} maps a coset.*

Proof. $\mathbf{u} \in C$ if and only if $\mathbf{u}H^T = \mathbf{0}$, thus the kernel of the map Φ is C. There are q^{n-k} cosets altogether, with each coset giving a unique syndrome and so maps onto $V(n-k,q)$. □

Build a column of vectors of $V(n,q)$ syndromes such that each v_i is in a distinct coset and v_i is a vector of minimal weight in its coset.

We just need to list of coset leaders and their corresponding syndromes to decode. With such a list, nearest neighbour decoding can be done as follows:

(1) Compute the syndrome $\mathbf{r}H^T$ of the received word $\mathbf{r} \in V(n,q)$.
(2) Find the coset leader $\mathbf{l}$ with the same syndrome.
(3) Decode to the codeword $\mathbf{r} - \mathbf{l}$.

This method is called *syndrome decoding*. Even syndrome decoding can be computationally cumbersome. Codes used in practice take advantage of specific properties to further refine the process of decoding.

Start with zero vector which has syndrome 0. Then take all vectors of weight 1 and compute their syndromes. Then take all vectors of weight 2 and compute their syndromes, and then keep doing this until all q^{n-k} unique syndromes have been found.

If v_1 has weight d_1 and v_2 has weight d_2 with $v_1H^T = v_2H^T$ then $v_1 - v_2 \in C$,

$$|v_1 - v_2| \leq |v_1| + |v_2| \leq d_1 + d_2$$

so either $v_1 = v_2$ or $d_1 + d_2 \geq d$ where $d(C) = d$.

This gives us an alternative method, stop generating syndromes after doing all vectors of weight less than $\left\lfloor \frac{d-1}{2} \right\rfloor$ since all syndromes are unique because d is the distance of the code. To decode v, compute vH^T if it is on the left as v_iH^T decode to $v - v_i$, otherwise more than $\left\lfloor \frac{d-1}{2} \right\rfloor$ errors and can't be uniquely decoded.

Incomplete decoding recognises these facts and uses a standard array to decode a linear code C with $d(C) = 2t+1$ or $d(C) = 2t+2$ as follows:

(1) If the received vector $\mathbf{y}$ is in a row whose coset leader has weight $\leq t$, assume at most t errors occurred and decode as usual.
(2) if the received vector $\mathbf{y}$ is in a row whose coset leader has weight $> t$, then more than t errors have occurred and request for retransmission.

Thus we correct all instances of at most t errors, and detect some instances of more than t errors. Of course, such a request may be time consuming or impossible.

6. Coding Bounds

In this section we are back to discussing more general codes. They do *not* have to be linear.

An (n, M, d)-*code* is a code of length n containing M codewords and having distance d.

- Note the difference between an (n, M, d)-code and an $[n, k, d]$-code. The former makes sense for all codes where codewords have fixed length, the latter makes sense only for linear codes.
- An $[n, k, d]$-binary code is an $\big(n, 2^k, d\big)$-code.

Large information rate (low redundancy) and large minimal distance are often trade-offs for one another.

How do we determine whether one code is better than another? Since there are three variables, it can sometimes be hard to make comparisons. But if two of the variables are fixed, then codes *can* be compared. In this section, we will consider possible codes that can arise when n and d are fixed and M is allowed to vary. Clearly, an (n, M_1, d)-code is better than an (n, M_2, d)-code precisely if $M_1 > M_2$. We then ask the following:

- What is the maximum number of codewords a code of fixed distance d and length n can have?

Definition 6.1. Denote by $A_q(n, d)$ the largest value of M such that there exists an (n, M, d)- code over a q-ary alphabet $\mathrm{GF}(q)$.

Example 6.2. $A_2(5, 3) \geq 4$ since

$$C = \{00000, 11100, 00111, 11011\}$$

is a linear binary code of length 5, distance 3 and dimension 2 with four codewords.

Another example is $A_q(n, 1)$. Trivially, $V(n, q)$ is a code of length n and distance 1 and so $M = q^n$.

Lastly, take $A_q(n, n)$. Since minimal distance must n and there are n positions, we must have each codeword be made up of different symbols. A simple example would be $A_q(n, n) = q$ since we can have a codeword of length n and made up of a single element in $\mathrm{GF}(q)$.

Recall that for positive integers n and m, the *binomial coefficient* $\binom{n}{m}$ (read "n choose m") is defined by

$$\binom{n}{m} = \frac{n!}{m!(n-m)!}.$$

Lemma 6.3. *The number of m-element subsets of an n-element set is given by the binomial coefficient* $\binom{n}{m}$.

Proof. Let X have cardinality n. We get $\varnothing \in \mathcal{P}(X)$ is the only subset of X with cardinality 0. Furthermore,

$$\binom{n}{0} = \frac{n!}{0!(n-0)!} = 1$$

and so the base case holds. Suppose that there exists an $m < n$ such that the number of m-element subsets of X is given by the binomial coefficient $\binom{n}{m}$. Suppose M is an m-element subset of X. We can then construct an $m+1$-element subset of X by taking each of the remaining $n - m$ elements of X that are not in M and adding them to M. Yet, by doing so, we will repeat sets that previously differed by 1 element. This will happen for each element added, and thus we must divide by $m + 1$. This gives us the total number of $m + 1$-element subsets of X as

$$\frac{n!}{(m+1)!(n-(m+1))!}$$

and so by induction the hypothesis holds. □

Definition 6.4. Let $V(n,q)$ denote the set of all ordered n-tuples with each entry in GF(q). For a vector $\mathbf{u} \in V(n,q)$, and a non-negative integer r, the *sphere* of radius r and centre $\mathbf{u}$ is the set

$$S(\mathbf{u},r) = \{\mathbf{v} \in V(n,q) \mid d(\mathbf{u},\mathbf{v}) \leq r\}.$$

Note. If we take $r = \lfloor \frac{d-1}{2} \rfloor$ then the spheres will be disjoint.

Lemma 6.5. *A sphere of radius r in $V(n,q)$ contained precisely*

$$\sum_{i \leq r} \binom{n}{i} (q-1)^i$$

vectors.

Proof. First note that $|S(\mathbf{u},r)| = |S(\mathbf{0},r)|$. The map from $S(\mathbf{u},r)$ onto $S(\mathbf{0},r)$ defined $\mathbf{v} \mapsto \mathbf{v} - \mathbf{u}$ is a bijection. We get

$$d_H(\mathbf{u},\mathbf{v}) \leq r \Longleftrightarrow d_H(\mathbf{v}-\mathbf{u},\mathbf{0}) \leq r.$$

Also,

$$\begin{aligned}
|S(\mathbf{0},r)| &= |\{\mathbf{v} \mid w(\mathbf{v}) \leq r\}| \\
&= |\{\mathbf{v} \mid w(\mathbf{v}) = 0\}| + |\{\mathbf{v} \mid w(\mathbf{v}) = 1\}| + \ldots + |\{\mathbf{v} \mid w(\mathbf{v}) = r\}| \\
&= 1 + n(q-1) + \frac{n(n-1)}{2}(q-1)^2 + \ldots + \binom{n}{r}(q-1)^r \\
&= \sum_{i \leq r} \binom{n}{i} (q-1)^i
\end{aligned}$$

□

Theorem 6.6. *A q-ary $(n, M, 2t+1)$-code satisfies*

$$M[\sum_{i \leq t} \binom{n}{i} (q-1)^i] \leq q^n.$$

Proof. Note that if $c_1 \neq c_2 \in C$, then their spheres are disjoint for $t = \lfloor \frac{d-1}{2} \rfloor$. Then

$$|\cup_{c \in C} S(c,t)| \leq q^n$$

and is disjoint so

$$\sum_{c \in C} |S(c,t)| = M|S(\mathbf{0},t)| \leq q^n$$

and $S(\mathbf{0},t) = \sum_{i \leq t} \binom{n}{i} (q-1)^i$. □

Note. It immediately follows in an $(n, M, 2t+1)$-code,

$$M \leq \frac{q^n}{\sum_{i \leq r} \binom{n}{i} (q-1)^i}.$$

Thus the Hamming bound is an *upper bound* for the number of codewords in a code with a given length n and odd distance $2t+1$. For $q = 2$, the Hamming bound reduces to

$$M[\binom{n}{0} + \binom{n}{1} + \ldots + \binom{n}{t}] \leq 2^n.$$

For codes with even distance $2t+2$, $t = \lfloor \frac{d-1}{2} \rfloor = \lfloor t + \frac{1}{2} \rfloor$.

Theorem 6.7. *A q-ary (n, M, d)-code C satisfies*

$$A_q(n, d) \leq q^{n-d+1}. \tag{1}$$

Proof. Consider the map $C \to A^{n+1-d}$, by getting rid of last $d-1$ coordinates. We claim that this is one-to-one. If c_1, c_2 only differ in the last $d-1$ coordinates, so $d(c_1, c_2) \leq d-1$ which implies $c_1 = c_2$ since $d(C) = d$. Also, $|C| \leq \left|A^{n+1-d}\right| = |A|^{n+1-d}$ and so $M \leq q^{n+1-d}$ proving (1). □

Note. MDS codes reach the bound (1).

Corollary 6.8. *For any $[n, k, d]$-linear code*

$$k \leq n - d + 1.$$

Proof. We get that $M = q^k$ and we know $M \leq q^{n-d+1}$ and so it follows that $k \leq n - d + 1$. □

Theorem 6.9.

$$A_q(n, d) \geq \frac{q^n}{\sum_{i<d} \binom{n}{i} (q-1)^i}$$

Proof. Suppose we found $c_1, \ldots, c_N \in V(n, q)$ such that $d(c_i, c_j) \geq d$ for all $i \neq j$. Then the size of the union of their spheres with $d-1$ must be determined. We get that

$$\sum_{i=1}^{N} |S(c_i, d-1)| \leq N \sum_{j<d} \binom{n}{j} (q-1)^j = N|S(\mathbf{0}, d-1)|.$$

If

$$N|S(\mathbf{0}, d-1)| < q^n \tag{2}$$

there exists $c \in V(n, q)$ such that $c \notin \bigcup_{i=1}^{N} S(c_i, d-1)$. So we assumed that $\{c_1, \ldots, c_N\}$ is a $(n, N+1, d)$-code. Conclusion, we can increase the code while condition (2) holds so we stop when it fails. So

$$M \geq \frac{q^n}{|S(\mathbf{0}, d-1)|}$$

getting a code of size M. □

Theorem 6.10. *There are also linear $[n, k, d]$-codes with*

$$q^k \geq \frac{q^n}{|S(\mathbf{0}, d-1)|}.$$

Proof. We want the code to be *linear*. If the original code is C_0 and we want to add c to C_0 we need to add all $\lambda c + c_0$ with $c_0 \in C_0$ and $\lambda \in \mathrm{GF}(q)$. We want to make sure that the weight of $\lambda c + c_0$ is greater than or equal to d, unless it is zero. If $\lambda \neq 0$,

$$w(\lambda c + c_0) = w\big(c + \lambda^{-1} c_0\big) = d\big(c, -\lambda^{-1} c_0\big)$$

but if $c_0 \in C_0$, then $-\lambda^{-1} c_0 \in C_0$ so their distance is greater than or equal to 0. If $\lambda = 0$, then $w(c_0) \geq d$ if $c_0, c \neq 0$. As long as $|S(0, d-1)| < q^n$, increase the code to desired bound. □

7. Perfect Codes

Definition 7.1. A q-ary $(n, M, 2t+1)$-code is *perfect* if it achieves the Hamming bound, that is, if

$$M = \frac{q^n}{\binom{n}{0} + \binom{n}{1}(q-1) + \ldots + \binom{n}{t}(q-1)^t}.$$

Thus, a perfect t-error correcting code is one for which the M spheres of radius t centred on the M codewords fill the whole space $V(n,q)$ without overlapping, that is, every vector in $V(n,q)$ is at distance at most t from exactly one codeword (refer to Theorem 6.6).

Example 7.2. Here we will give two examples, called the *trivial* perfect codes. They are when $C = V(n,q)$ such that $M = q^n$ and $C = \{\mathbf{0}, \mathbf{1}\}$ such that $M = 2$. Both meet the Hamming bound.

Definition 7.3. Let r be a positive integer and let H be a $r \times (2^r - 1)$ matrix whose columns are the distinct non-zero vectors of $V(r,2)$. Then the *linear* code having H as its parity-check matrix is called a *binary Hamming code* and is denoted by $\mathrm{Ham}(r,2)$.

Note. We get that

- $\mathrm{Ham}(r,2)$ has length $n = 2^r - 1$ and dimension $k = n - r$. Thus, $r = n - k$ is the number of symbols added when encoding, that is, r is the *redundancy* of the code.
- The columns of H may be written in any order, so $\mathrm{Ham}(r,2)$ is really any one of a number of equivalent codes. This does not cause a problem.

Theorem 7.4. *For all positive integers $r \geq 2$, the binary Hamming code* $\mathrm{Ham}(r,2)$ *is a $[2^r - 1, 2^r - 1 - r, 3]$-linear code. Moreover,* $\mathrm{Ham}(r,2)$ *is perfect.*

Proof. We get that $M = |C| = 2^{2^r - 1 - r}$ and $d = 3$, which implies $t = 1$. The Hamming bound gives us

$$\frac{2^{2^r-1}}{1 + 2^r - 1} = 2^{2^r - 1 - r} = M$$

and so $\mathrm{Ham}(r,2)$ is perfect.

An alternative proof: Since columns of H are all distinct non-zero vectors of length m, we get all syndromes are columns of H. Standard array will have coset leaders of weight 0 and 1 only. So, every given length n vector can be corrected by changing at most one coordinate. Then for all vectors in $V(n,2)$, there exists a codeword in $\mathrm{Ham}(r,2)$, the distance is ≤ 1. So, $\bigcup_{c \in C} s(c,1) = V(n,2)$ which means $\mathrm{Ham}(r,2)$ is perfect. □

Note. Since $\mathrm{Ham}(r,2)$ is a perfect single-error correcting code, the coset leaders are precisely the 2^r vectors of weight at most one.

Now, the syndrome of the coset leader $0\ldots010\ldots0$ with 1 is the j-th position is $(0\ldots010\ldots0)H^T$ which is the transpose of the j-th column of H. Hence, by arranging the columns of H in order of increasing binary numbers so that the j-th column of H is the binary representation of j, nearest neighbour decoding can be done as follows:

(1) Compute the syndrome $\mathbf{r}H^T$ of the received word $\mathbf{r}$.
(2) If $\mathbf{r}H^T = \mathbf{0}$, decode to $\mathbf{r}$.
(3) If $\mathbf{r}H^T \neq \mathbf{0}$, the vector $\mathbf{r}H^T$ gives the binary representation of the error position (assuming a single error).

Theorem 7.5. (van Lint and Tietavainen, 1973) *A non-trivial perfect q-ary code, where q is a prime power, must have the same parameters as one of the Hamming or Golay codes.*

Note that the focus of Coding Theory is more on good decoding algorithms as opposed to good parameters.

7.1. **q-ary Hamming Codes.** We have seen Hamming codes for binary codes. We now wish to generalise this over an arbitrary q-ary alphabet. That is, we are working over $\mathrm{GF}(q)$ with distance 3 by constructing a parity-check matrix with r rows with as many columns as possible (keeping distance 3). Distance 3 gives us the columns are non-zero and no two columns are linearly dependent. There are three linearly dependent columns.

Recall that $h_1 \neq \mathbf{0}, h_2 \neq \mathbf{0}$ are linearly dependent if there exists non-zero $a, b \in \mathrm{GF}(q)$ such that $ah_1 + bh_2 = 0 \iff h_1 = -a^{-1}bh_2$.

Definition 7.6. Let r be a positive integer and let H be a $r \times \left(\frac{q^r-1}{q-1}\right)$ matrix whose columns are the distinct non-zero vectors of $V(n,q)$. Then the code having H as its parity-check matrix is called a *q-ary Hamming code* and is denoted by $\mathrm{Ham}(r,q)$.

Theorem 7.7. *For all positive integers $r \geq 2$, the q-ary Hamming code* $\mathrm{Ham}(r,q)$ *is a $[\frac{q^r-1}{q-1}, \frac{q^r-1}{q-1} - r, 3]$ code. Furthermore,* $\mathrm{Ham}(r,q)$ *is perfect.*

Proof. Let $d = 3$ and H be the parity-check matrix. We get that the columns must be non-zero and not multiples of each other. Hence, for each column of weight j we must choose j elements and can only change $(q-1)^j$ many. This corresponds to

$$\sum_{j=1}^{r} \binom{r}{j} (q-1)^{j-1}$$

many columns. We make note that

$$\sum_{j=1}^{r} \binom{r}{j} (q-1)^{j-1} = \frac{1}{q-1}\left(-1 + \sum_{j=0}^{r} \binom{r}{j} (q-1)^j\right) = \frac{q^r-1}{q-1}$$

and so we get the desired length. The dimension is

$$\frac{q^r-1}{q-1} - r$$

and so $\mathrm{Ham}(r,q)$ is a $[\frac{q^r-1}{q-1}, \frac{q^r-1}{q-1} - r, 3]$ code.

Now, the cardinality of C is $q^k = q^{\frac{q^r-1}{q-1} - r}$. Since

$$\frac{q^n}{|S(\mathbf{0},1)|} = \frac{q^n}{1+q^r-1} = q^{n-r} = |C|,$$

the code is perfect. □

8. Reed-Solomon Codes

Definition 8.1. A *Vandermonde matrix* is an $n \times n$ matrix of the form $V = [a_i^j]_{i=1,\ldots,n, j=0,\ldots,n-1}$ where $a_1, \ldots, a_n$ are elements of some field F. That is,

$$V = \begin{bmatrix} 1 & 1 & \ldots & 1 \\ a_1 & a_2 & \ldots & a_n \\ a_1^2 & a_2^2 & \ldots & a_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{n-1} & a_2^{n-1} & \ldots & a_n^{n-1} \end{bmatrix}.$$

Lemma 8.2. *Let $\mathbb{F}$ be a field and $a_1, \ldots, a_n \in \mathbb{F}$ be distinct. Let $D = [a_{ij}]$*

$$a_{ij} = \begin{cases} a_i & \text{if } i = j; \\ 0 & \text{if } i \neq j. \end{cases}$$

be a diagonal $n \times n$ matrix over $\mathbb{F}$. Then

$$\mathbb{F}^n = \operatorname{span}\big(D^0\mathbf{1}, D^1\mathbf{1}, \ldots, D^{n-1}\mathbf{1}\big)$$

where $\mathbf{1} \in \mathbb{F}^n$ is the vector with all entries $1 \in \mathbb{F}$ (the multiplicative identity).

Proof. Note that

$$\mathbb{F}^n = \operatorname{span}\big(D^0\mathbf{1}, D^1\mathbf{1}, \ldots, D^{n-1}\mathbf{1}\big)$$

if and only if $D^0\mathbf{1}, \ldots, D^{n-1}\mathbf{1}$ are linearly independent. We will give a precise definition for each $D^k\mathbf{1}$ to demonstrate that they are indeed linearly independent. To do so, we need the standard basis of $\mathbb{F}^n$. We get that

$$D^k\mathbf{1} = \sum_{i=1}^{n} a_i^k \mathbf{e}_i.$$

It therefore follows that

$$b_0 D^0\mathbf{1} + b_1 D^1\mathbf{1} + \ldots + b_{n-1} D^{n-1}\mathbf{1} = \mathbf{0}$$

if and only if

$$\sum_{k<n} b_n \sum_{i=1}^{n} a_i^k \mathbf{e}_i = \mathbf{0}.$$

We will show that the constants $b_0, \ldots, b_{n-1} \in \mathbb{F}$ must be equivalent to 0 by deriving a contradiction. So, we assume linear dependency. Observe that

$$\sum_{k<n} b_n \sum_{i=1}^{n} a_i^k \mathbf{e}_i = \mathbf{0}.$$

$$\Rightarrow \qquad \sum_{k<n} b_k a_i^k \mathbf{e}_i = \mathbf{0}$$

for all $i \in \mathbb{N} : 0 < i \leq n$. This implies that

$$\sum_{k<n} b_k a_i^k = 0$$

for every $i \in \mathbb{N} : 0 < i \leq n$, which in turn implies that each $b_k = 0$ thereby giving us a contradiction. The reason for why it implies $b_k = 0$ is as follows. We define a function $g : \mathbb{F} \to \mathbb{F}$ as

$$g(x) = \sum_{k<n} b_k x^k.$$

We then have n many distinct roots for the polynomial $g(x)$ of degree $(n-1)$ which implies that it is the zero-polynomial. That is to say, every $b_k = 0$ and thus $D^0\mathbf{1}, \ldots D^{n-1}\mathbf{1}$ are linearly independent as required. □

Theorem 8.3. *The determinant of the Vandermonde matrix V is*

$$\det(V)(-1)^{n(n-1)/2} \prod_{i<j} (a_i - a_j).$$

Proof. We will provide a proof by induction on n. We get that the base case of $n = 2$ holds. So, we get

$$\det \begin{bmatrix} 1 & 1 & \dots & 1 & 1 \\ a_1 & a_2 & \dots & a_n & x \\ a_1^2 & a_2^2 & \dots & a_n^2 & x^2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_1^{n-1} & a_2^{n-1} & \dots & a_n^{n-1} & x^{n-1} \end{bmatrix} = f(x)$$

of degree $(n-1)$. If we have $x = a_i$ for some $i = 1, \dots, n-1$ then the last column is the same as another and so $\det(V) = 0$. Thus, for $x = a_{n+1}$,

$$\begin{aligned} f(x) &= \det(V)(x - a_1)\dots(x - a_{n-1}) \\ &= (-1)^{n(n-1)/2} \prod_{i<j} (a_i - a_j)(x - a_1)\dots(x - a_{n-1}) \\ &= (-1)^{n(n+1)/2} \prod_{i<j} (a_i - a_j) \end{aligned}$$

and so the hypothesis holds for all $n \geq 2$. □

Note. In particular, if $a_1, \dots, a_n$ are distinct, then the Vandermonde matrix is invertible.

Definition 8.4. Consider now a code C with parity check matrix $H = [a_j^i]_{i=0,\dots,n-k-1, j=1,\dots,n}$, for some k, with $a_1, \dots, a_n$ distinct. Since the $(n-k) \times (n-k)$ minors are Vandermonde, they all have non-zero determinant so C has minimum distance $d = n-k+1$ by Theorem 4.10. They are MDS codes, i.e. attain the Singleton bound (Corollary 6.8).

The condition that $a_1, \dots, a_n$ are distinct elements of $\mathrm{GF}(q)$ forces $n \leq q$ so the length of these codes are restricted by the size of the field. We will look at the case $n = q$, so $\{a_1, \dots, a_q\} = \mathrm{GF}(q)$. They are called *Reed-Solomon codes* denoted $RS_k(q)$.

Note. The 2D barcodes (QR codes) that are used by the NZ government Covid Tracer app use Reed-Solomon codes over GF(256) for error correction.

Example 8.5. Let $H_k = [a_j^i]_{i=0,\dots,q-k-1, j=1,\dots,q}$ for $\{a_1, \dots, a_q\} = \mathrm{GF}(q)$. We will look at an example with $q = 5$ and $k = 2$. Then

$$H = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 4 & 4 & 1 \end{bmatrix}$$

is a $[5, 2, 4]$-code with the generator matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix}.$$

Theorem 8.6. *The generator matrix of $RS_k(q)$ is H_{q-k}. So the dual of $RS_k(q)$ is $RS_{q-k}(q)$.*

Proof. We get that

$$\left(a_1^i, \dots, a_q^i\right) \cdot \left(a_1^j, \dots, a_q^j\right) = a_1^{i+j} + \dots + a_q^{i+j}$$

which implies that $i \leq q-k-1$ and $j \leq q-(q-k)-1 = k-1$ and so $i+j \leq q-2 < q-1$. We need to prove that if $m < q-1$, then $S_m = a_1^m + \dots + a_q^m = 0$ if $\{a_1, \dots, a_q\} = \mathrm{GF}(q)$.

We will use the fact that a polynomial of degree m has at most m roots. So the polynomial $x(x^m - 1)$ has at most $m+1$ roots. Let $b \in \mathrm{GF}(q)$ where $b \neq 0$ such that $b^m - 1 \neq 0$. This exists if $m + 1 < q$. Note then that

$$\mathrm{GF}(q) = \{a_1, \ldots, a_1\} = \{ba_1, \ldots, ba_q\}.$$

So, $S_m = a_1^m + \ldots + a_q^m = (ba_1)^m + \ldots + (ba_q)^m$ which gives us that $S_m = b^m S_m$. If $S_m \neq 0$, it implies that $b^m - 1 = 0$ (a contradiction). Thus, $S_m = 0$ which proves the result. □

For Reed-Solomon codes, they are $[q, k, q-k+1]$-codes that achieve the Singleton bound since $k + d = n + 1$ for $d = q - k + 1$ and $n = q$ over $\mathrm{GF}(q) = \{a_1, \ldots, a_q\}$. For q prime, we can take $\mathrm{GF}(q) = \{0, 1, \ldots, q-1\}$. The $k \times q$ generator matrix is

$$G = \begin{bmatrix} 1 & 1 & \ldots & 1 \\ 0 & 1 & \ldots & q-1 \\ 0^2 & 1^2 & \ldots & (q-1)^2 \\ \vdots & \vdots & \ddots & \vdots \\ 0^{k-1} & 1^{k-1} & \ldots & (q-1)^{k-1} \end{bmatrix}$$

and the $(n-k) \times q$ parity-check matrix is

$$H = \begin{bmatrix} 1 & 1 & \ldots & 1 \\ 0 & 1 & \ldots & q-1 \\ 0^2 & 1^2 & \ldots & (q-1)^2 \\ \vdots & \vdots & \ddots & \vdots \\ 0^{q-k-1} & 1^{q-k-1} & \ldots & (q-1)^{q-k-1} \end{bmatrix}.$$

If we add $k-1$ and $q-k-1$ we get $q-2$, so it follows that GH^T is the zero matrix.

In erasure decoding we are interested in the situation in which some coordinates of a received codeword cannot be read, instead of being read incorrectly. One example is when part of a QR code is obstructed or missing. Reed-Solomon codes are very good in the context of erasure decoding.

Theorem 8.7. *If we know any k coordinates of a codeword of $RS_k(q)$, then we can recover the other $q-k$ coordinates and the message.*

Proof. Suppose we have the message $M = (m_0, \ldots, m_{k-1})$ and encode by MG for generator matrix

$$G = \begin{bmatrix} 1 & 1 & \ldots & 1 \\ a_1 & a_2 & \ldots & a_q \\ a_1^2 & a_2^2 & \ldots & a_q^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{k-1} & a_2^{k-1} & \ldots & a_q^{k-1} \end{bmatrix}$$

which gives us $c_j = \sum_{i<k} m_i a_j^i$ for each $1 \leq j \leq q$. Suppose we know k coordinates of the codeword which is a subset (of cardinality k) of $\{c_1, \ldots, c_q\}$. The determinant will be non-zero since the k columns are LI (it is a Vandermonde matrix). We can then recover the message since the Vandermonde matrix is invertible and so we can use this to find the other $q-k$ coordinates. □

A different application of this result is the Shamir secret sharing scheme. We want to distribute some secret (say a password) among q (or fewer) different users so that any k users can recover the secret but $k-1$ users cannot. This prevents a small group from obtaining the secret and guarantees cooperation. If our secret is a message $(m_1, \ldots, m_k)$ from $V(k, q)$ then, by the above, we can encode it as a

codeword of $RS_k(q)$ and give one coordinate to each user. We see, as above, that k users can recover the secret. We can also see that $k-1$ coordinates is not enough.

Another use of the same set of ideas is for file comparison. Suppose you need to compare two large files at different locations (e.g. to see if there is a new version needed to backup) but want to avoid sending through the whole file unless necessary. Assume the files are represented as elements of $V(k,q)$ for suitable k,q. On one location, the user chooses a random i between 1 and q, computes the i-th coordinate of the encoding of the file as a codeword of $RS_k(q)$ and sends i and the corresponding coordinate to another location. The user at the second location then computes the i-th coordinate received from the first user. If the values of the i-th coordinate differ, the files differ. What is the chance that the i-th coordinates are the same but the files differ? Since the code has distance $q-k+1$, the coordinate can only be the same in at most k places and i was chosen at random so there is only a $\frac{k}{q}$ chance (or less) of this happening. By choosing q much larger than k, we can make this a very small probability.